AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 6 · STUDENT LESSON

Sound Becomes a Signal

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 3: comparisons need shared sensor settings.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Interpret an ordered sequence of sound-signal samples with a stated time interval. Try the recall before opening notes.

Your learning target

I can interpret an ordered sequence of sound-**signal sample**s with a stated time interval.

I can explain why summaries or sparse **sample**s can omit changes.

I can separate an audio recording, a **transcript** and an interpretation of meaning.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Computers receive changing measurements, not human-like hearing.



Your target

LEARNING TARGET A whisper recorded close to a microphone has larger **sample amplitude**s than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

Learner name:

Readiness for the connected explanation

Two image grids have the same number of black pixels. Why can their pictures differ?

This week's end goal

By the end, I can connect ordered sound **sample**s to timing and explain what a maximum loses.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Sound Becomes a Signal case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A microphone converts air-pressure changes into a sequence of measured values that can be plotted as a **waveform**.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Sound involves changing pressure in a medium such as air. A microphone responds to that variation and produces a changing **signal**. A digital recording stores measurements of the **signal** at successive times.

The values are ordered: swapping their times changes the represented pattern.

We will use tiny invented sequences to study this idea on paper. Their values are arbitrary **signal** units, not decibels or a test of safe listening levels.

Real audio uses many more **sample**s, so our five-point examples should not be played as if they were a realistic recording of speech.

A recording must include enough information to interpret the values, including their time spacing and encoding. The sequence 0,2,0,2,0 does not by itself say how quickly the changes happened.

This is similar to needing image dimensions as well as pixel values.

STUDENT LESSON · 3 OF 18

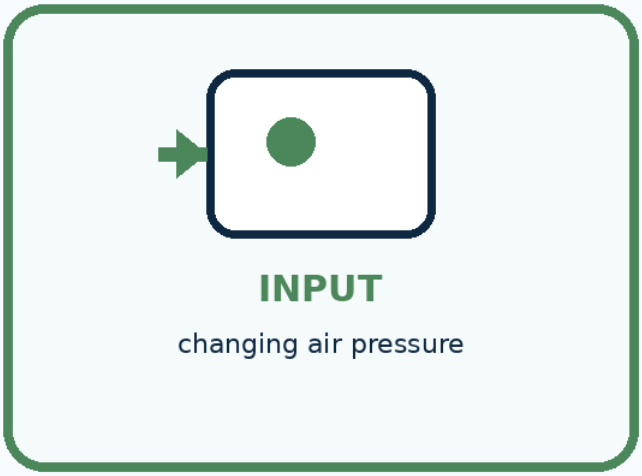
See how it works

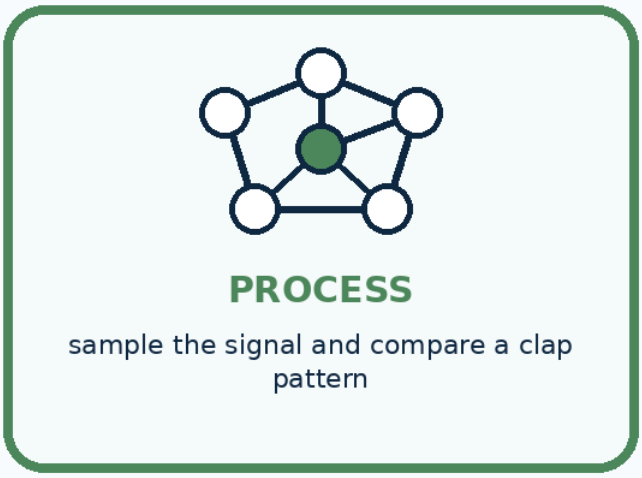
> Follow the mechanism for Sound Becomes a Signal. Connect each labelled input, deciding step and result.

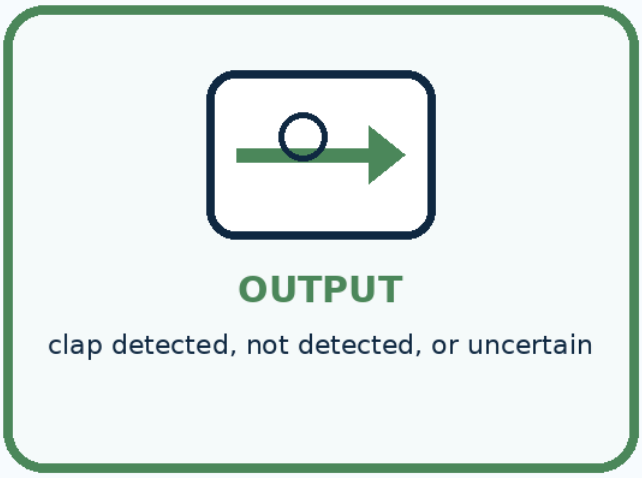
VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT changing air pressure

PROCESS sample the **signal** and compare a clap pattern

OUTPUT clap detected, not detected, or uncertain

Key idea

CORE IDEA Computers receive changing measurements, not human-like hearing.

How the mechanism works

UNDER THE HOOD Audio is represented by **sample**s taken over time. **Sample** rate describes how often measurements are taken; **amplitude** describes **signal** strength.

Noise can overlap the target pattern, so detection requires a decision rule or model and an uncertainty policy.

Draw a fresh example with all three stages:

A closer explanation

Three stages in a speech system

INPUT	OPERATION	OUTPUT
Sound reaching a microphone	Measure the changing signal at successive times	Ordered samples and timing information
Recorded audio	Speech recognition estimates the words	A transcript that may contain errors
Transcript and an explicit request	A separate rule or model processes the text	An answer or proposed action

Recording, transcribing and interpreting are separate operations. A later stage can correctly process an incorrectly transcribed number.

Time in milliseconds · 0 · 1 · 2 · 3 · 4

> **Time in milliseconds:** Invented signal value
>
> **0:** 0

> **1:** 2
>
> **2:** 0
>
> **3:** 2
>
> **4:** 0

There are five **sample**s with four gaps between the first and last. Each gap is 1 millisecond, so the elapsed time from first to last is 4 milliseconds.

If the same five values were spaced 2 milliseconds apart, they would span 8 milliseconds. Equal value lists with different timing do not describe the same time pattern.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

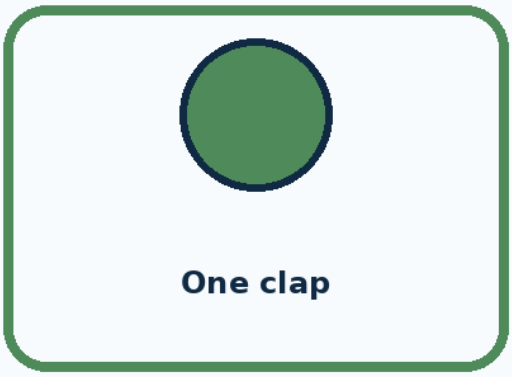
Compare the cases

> Compare the supplied cases for Sound Becomes a Signal. Identify the detail that changes the conclusion.

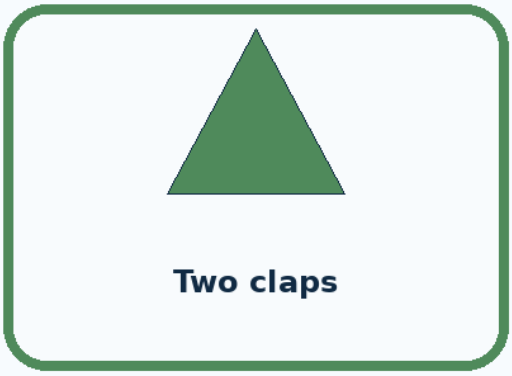
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.



One clap



Two claps



Talking



The case to investigate

CASE 1 — One clap Clue: single sharp peak. Meaning: target pattern.

The case to investigate

CASE 2 — Two claps Clue: two separated peaks. Meaning: different timing pattern.

The case to investigate

CASE 3 — Talking Clue: many irregular changes. Meaning: possible false trigger.

The case to investigate

CASE 4 — Quiet room Clue: small background values. Meaning: baseline.

Keep this boundary clear

BOUNDARY A microphone measures a **signal**; it does not understand a speaker's meaning by itself.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Retain only **sample**s at 0,2,4 ms.

Predicted result

Prediction: Both retained sequences are 0,0,0.

Why the result follows

Reason: The selected times omit both nonzero **sample**s from each **signal**.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Sound Becomes a Signal to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.


W
waveform


S
sample


A
amplitude



Word	Meaning	My example
waveform	graph of a signal changing over time	
sample	one measurement at one moment	
amplitude	size or strength of a signal value	
noise	unwanted signal mixed with the target	

Misconception detective

Find the unsupported claim

CLAIM TO REPAIR A loud sound must be a clap.

Use one vocabulary word and one piece of **evidence** to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Signal	A changing physical or recorded quantity that can carry information

Word	Meaning
Sampling interval	The time between successive recorded samples
Transcript	A text representation of spoken words, which may contain errors or omit sound details

STUDENT LESSON · 6 OF 18

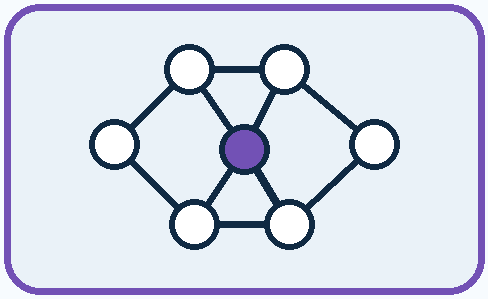
Follow a worked example

> Follow the worked trace for Sound Becomes a Signal, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE A microphone converts air-pressure changes into a sequence of measured values that can be plotted as a **waveform**.

Step 1

STEP 1 — OBSERVE changing air pressure

Step 2

STEP 2 — TRACE **sample** the **signal** and compare a clap pattern

Step 3

STEP 3 — CONCLUDE clap detected, not detected, or uncertain

Model answer

MODEL REASONING Look for two strong peaks separated by the expected gap. **Noise** may add extra peaks or hide a clap. A safe detector needs thresholds, timing rules, and a not-sure result when **evidence** is weak.

A claim to avoid

DO NOT OVERCLAIM A microphone measures a **signal**; it does not understand a speaker's meaning by itself.

What new **evidence** could change this answer?

A closer explanation

The largest value in 0,2,0,2,0 is 2. The largest value in 0,0,2,0,0 is also 2. The first sequence reaches 2 twice at different times, while the second does so once.

Keeping only the maximum loses that distinction, just as counting black pixels lost their arrangement.

Worked explanation · Sound Becomes a Signal

Sparse measurements can also miss changes. If we inspect the first sequence only at times 0, 2 and 4 milliseconds, all retained values are 0. The two intervening values of 2 disappear from the record.

It would be incorrect to conclude from these retained **sample**s alone that the **signal** stayed at zero between them.

Worked explanation · Sound Becomes a Signal

More frequent sampling can preserve faster changes under suitable recording conditions, but sampling frequency alone does not make every recording clear.

Microphone placement, background sound, clipping and the recording format can also matter. We are studying one source of missing information, not a complete audio engineering specification.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sound Becomes a Signal

A fictional microphone display records five **sample**s during a short classroom tap.

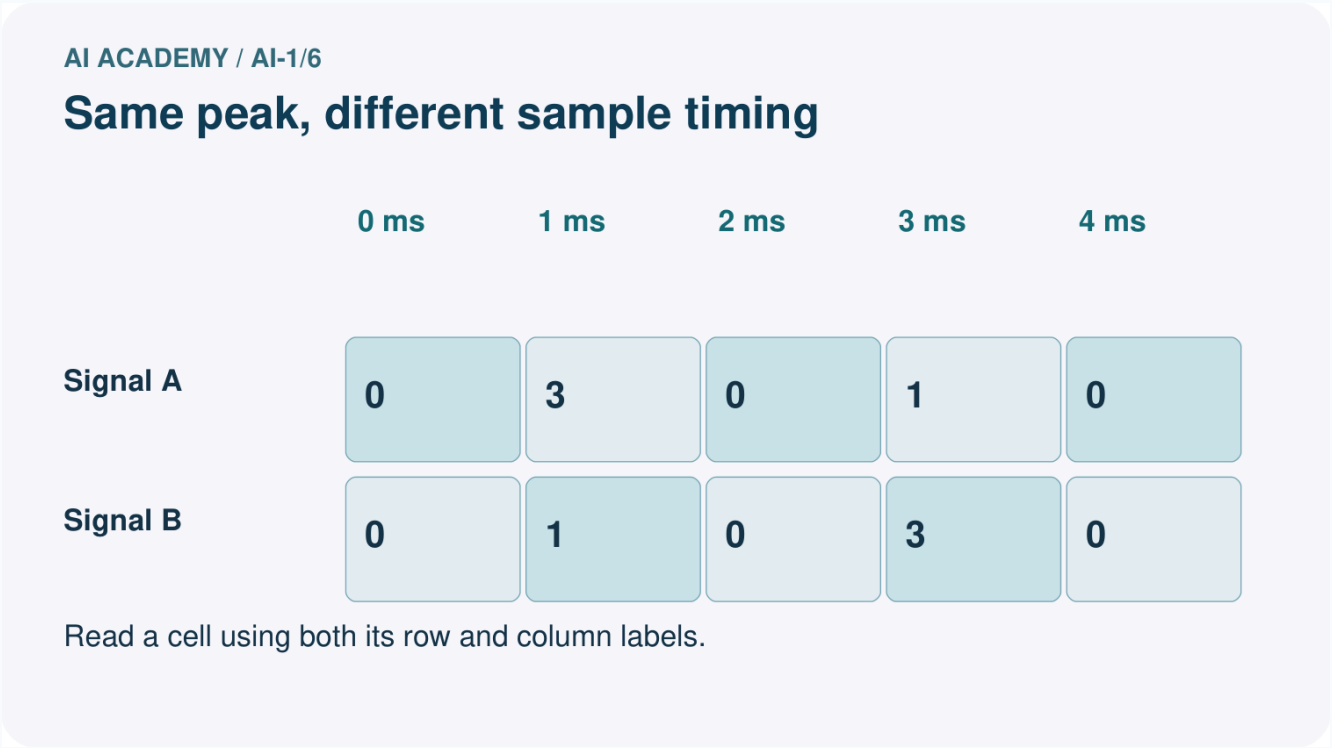
The facts we will use

Supplied facts

Times are 0,1,2,3,4 ms in order.

Supplied facts

Signal A: 0,3,0,1,0. **Signal** B: 0,1,0,3,0. Toy values are signed measurement units; these happen to be nonnegative.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/6

Sound Becomes a Signal

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Timed sampling

Associate every measurement with its sampling time.



2. Signal A

Its largest sample 3 occurs at 1 ms.



3. Signal B

Its largest sample 3 occurs at 3 ms.



4. Maximum summary

Both maxima are 3, but the strongest recorded tap arrives at different times.

CHANGE ONE THING

Retain only samples at 0,2,4 ms.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Timed sampling	Associate every measurement with its sampling time.
2. Signal A	Its largest sample 3 occurs at 1 ms.
3. Signal B	Its largest sample 3 occurs at 3 ms.
4. Maximum summary	Both maxima are 3, but the strongest recorded tap arrives at different times.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The sequences contain the same values in different orders. Keeping the maximum preserves peak recorded value while losing its timing.

A graph should keep time on the horizontal axis; shuffling **sample**s changes the recorded **signal** even if its maximum stays fixed.

What this example does not establish

Keep the stated limit

These sparse toy values cannot establish perceived loudness, pitch or what happened between sampling times.

Keep the stated limit

Read the labelled picture: Does the all-zero retained record prove the full **signal** was silent? Use the picture.

Keep the stated limit

Change one thing: Retain only **sample**s at 0,2,4 ms. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Sound Becomes a Signal claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
One clap	single sharp peak	target pattern
Two claps	two separated peaks	different timing pattern
Talking	many irregular changes	possible false trigger
Quiet room	small background values	baseline

Repair two tempting claims

Challenge claim A

CLAIM A A loud sound must be a clap.

Repair:

Challenge claim B

CLAIM B Silence means the sensor has no data.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

A **transcript** might record the words Fine, thank you while omitting a pause, a laugh or overlapping speech. It might also record the wrong words.

Text can be useful for searching spoken content, but it is not the same representation as the audio. Do not infer a speaker's emotions with certainty from a short **transcript**.

Check the claim against the facts

In a voice calculator, audio saying sixteen can be transcribed as sixty. A calculator can then correctly divide sixty by four and return fifteen. Inspecting the intermediate **transcript** identifies the first mismatch.

Confirming the interpreted numbers before acting can prevent this particular error from travelling into a later stage.

Explain one detail a transcript can preserve and one sound detail it may omit. Explain why a correct calculation does not prove the transcript was correct

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STUDENT LESSON · 8 OF 18

Practise together

Practise Sound Becomes a Signal with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Computers receive changing measurements, not human-like hearing.

Use a fresh example to explain **waveform**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Silence means the sensor has no data.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

A five-**sample** record has values 0,3,0,1,0 at times 0,2,4,6,8 milliseconds. Read the timing before answering. The first-to-last duration is the final time minus the initial time, not the **sample** count multiplied by the gap.

State the interval and first-to-last duration. If only times 0,4,8 are kept, list the retained values and identify a change the shorter record misses

To test a claim about the spaces between **sample**s, construct another possible sequence that agrees at retained times but differs between them.

If both match the stored record, that record alone cannot decide which intervening pattern occurred.

Practise with less help: Inspect what was discarded

Use workbook W2 for the connected example “Sound as **sample**d information”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Sound as **sample**d information. Proposed explanation: “Five **sample**s make five gaps”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Does the all-zero retained record prove the full **signal** was silent? Use the picture.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

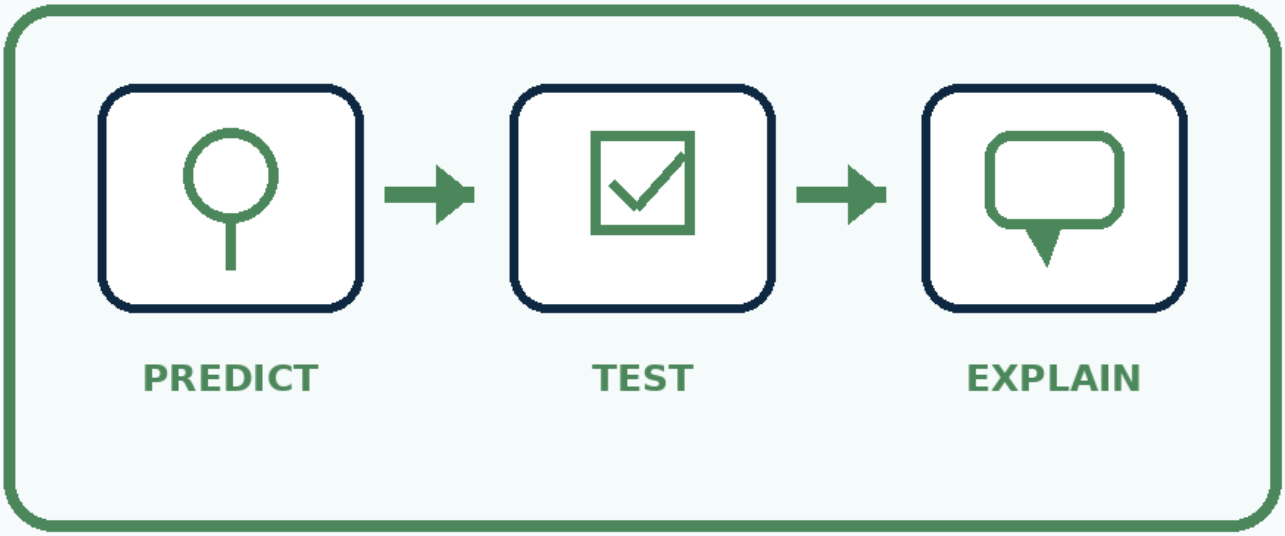
Run the investigation

> Investigate Sound Becomes a Signal. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Clap Pattern Detector

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Build a clap-pattern detector; test it with background **noise**.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

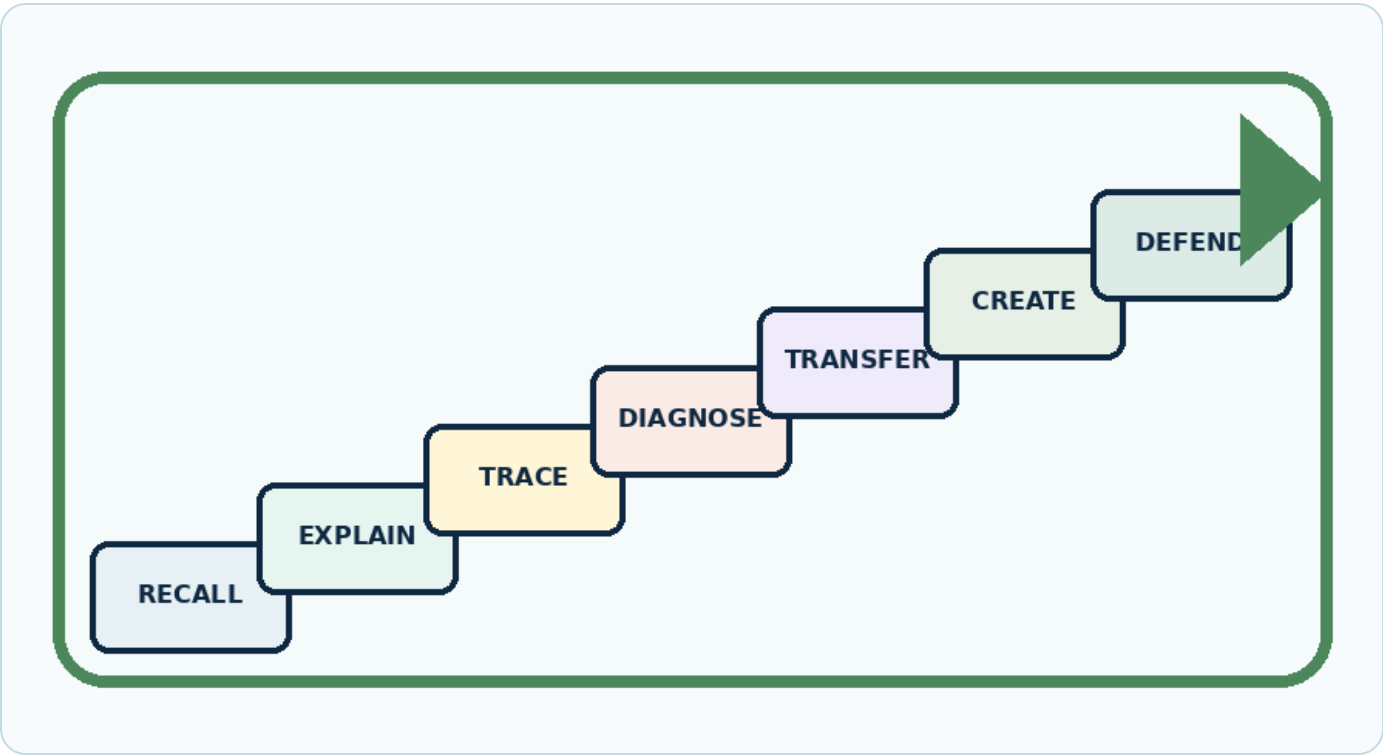
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Sound Becomes a Signal. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **waveform**.

EXPLAIN Computers receive changing measurements, not human-like hearing.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: A loud sound must be a clap.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

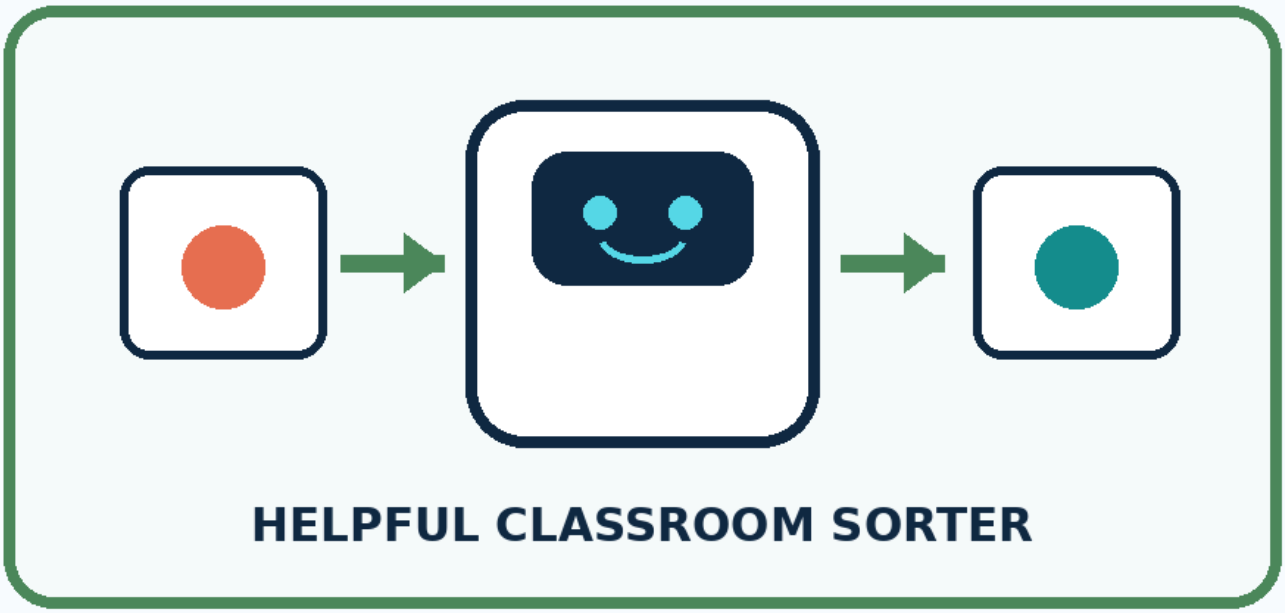
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT If the sorter uses sound, specify the exact sound pattern needed and test whether classroom **noise** makes that **input** unreliable.

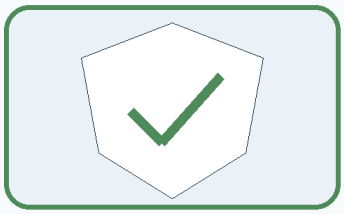
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN A microphone measures a **signal**; it does not understand a speaker's meaning by itself.

My project decision:

Evidence for your project

Record the sorter's **sampling interval** and retain relevant sequences when a decision depends on changes over time.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Use a **waveform** strip to decide whether a two-clap pattern occurred,

then explain how background **noise** could cause a missed or false detection.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Look for two strong peaks separated by the expected gap. **Noise** may add extra peaks or hide a clap. A safe detector needs thresholds, timing rules, and a not-sure result when **evidence** is weak.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Categories and Labels.

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Sound Becomes a Signal case for supported practice. Repeated examples do not become fresh assessments.

Week 6 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A microphone records quiet and loud claps at the same distance and settings.

The question

> What **waveform** difference is expected? Does it reveal a spoken word?

Worked explanation

The louder clap generally produces larger **amplitude**. **Amplitude** is not word meaning. Distance, gain and **noise** must be controlled.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Sound samples show changes in amplitude

Evidence or stage	Value or result
Quiet clap	1

Evidence or stage	Value or result
Loud clap	3

A microphone records quiet and loud claps at the same distance and settings.

> What **waveform** difference is expected? Does it reveal a spoken word?

The louder clap generally produces larger **amplitude**. **Amplitude** is not word meaning. Distance, gain and **noise** must be controlled.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Construct a new hidden change

A record keeps values 0 at times 0,3,6 milliseconds. Construct two seven-**sample** sequences for every whole-millisecond time from 0 through 6 that agree with those retained values but differ at omitted times.

Explain why the retained record cannot prove that nothing changed between measurements.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 6 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Two recordings contain the same word: one quiet and one loud. Which device receives the sound? Does a larger wave automatically mean a different word? Explain.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

If the sorter uses sound, specify the exact sound pattern needed and test whether classroom **noise** makes that **input** unreliable.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A whisper recorded close to a microphone has larger **sample amplitude**s than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Sound Becomes a Signal, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional microphone display records five **sample**s during a short classroom tap.

By the end, I can	How I will show it
Interpret an ordered sequence of sound-signal samples with a stated time interval	Explain the deciding step, show a trace or test, and state one limit.
Explain why summaries or sparse samples can omit changes	Explain the deciding step, show a trace or test, and state one limit.
Separate an audio recording, a transcript and an interpretation of meaning	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Sound Becomes a Signal without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Interpret an ordered sequence of sound-**signal sample**s with a stated time interval

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional microphone display records five **sample**s during a short classroom tap.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Both maxima are 3, but the strongest recorded tap arrives at different times.

1. **Signal** A

2. **Signal** B

3. Maximum summary

4. Timed sampling

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Maximum summary Both maxima are 3, but the strongest recorded tap arrives at different times. The sequences contain the same values in different orders.

Keeping the maximum preserves peak recorded value while losing its timing. A graph should keep time on the horizontal axis; shuffling **sample**s changes the recorded **signal** even if its maximum stays fixed.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Retain only **sample**s at 0,2,4 ms. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Both retained sequences are 0,0,0.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Both retained sequences are 0,0,0. The selected times omit both nonzero **sample**s from each **signal**.

Repair a missing building block

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention.

Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Review Sensors Are Messengers (ai-1/3). Goal: Describe a measurement using its quantity, value, unit and conditions. Short practice: Why can we calculate the earlier gap but not predict an exact new B reading?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

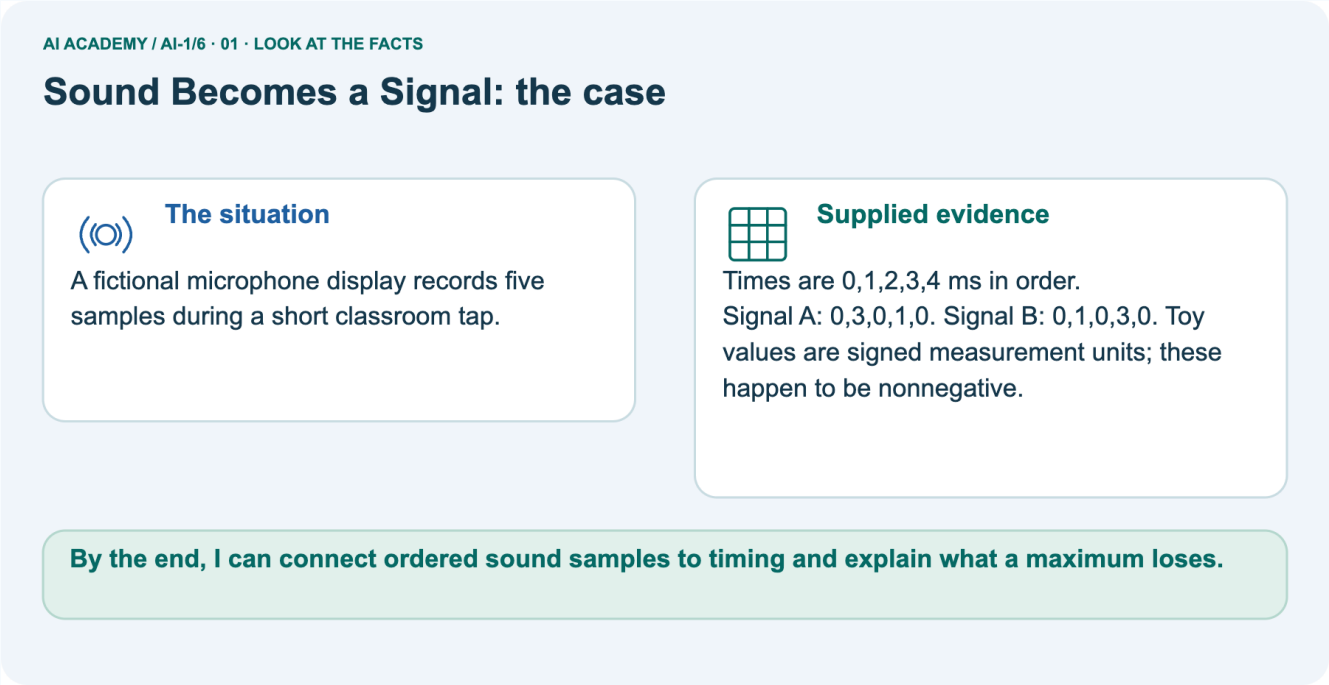
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Sound Becomes a Signal graphic using its labels, deciding evidence and relevant limitation.

01. Sound Becomes a Signal: the case



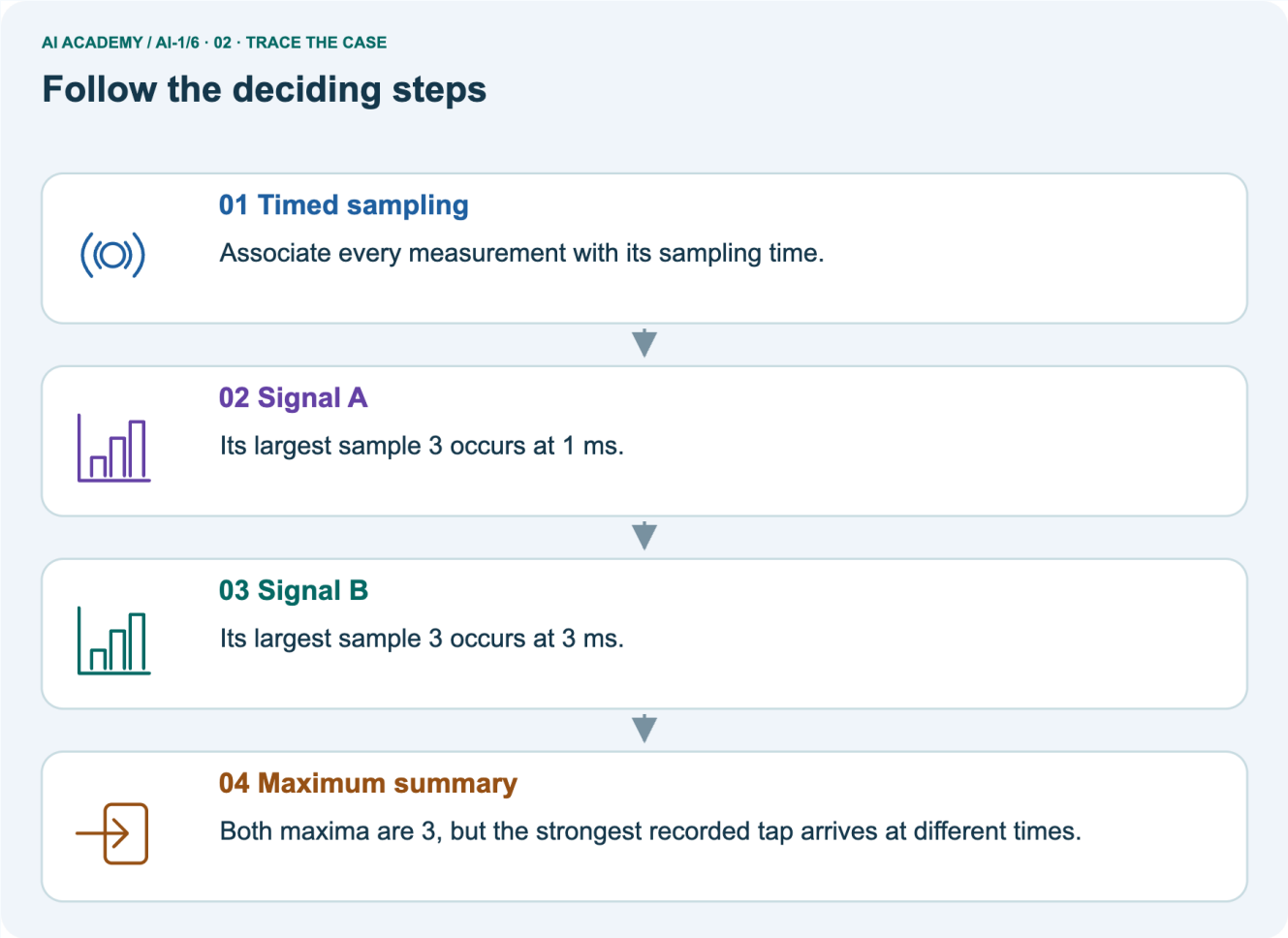
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional microphone display records five **sample**s during a short classroom tap. Supplied facts: Times are 0,1,2,3,4 ms in order. **Signal** A: 0,3,0,1,0. **Signal** B: 0,1,0,3,0.

Toy values are signed measurement units; these happen to be nonnegative.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

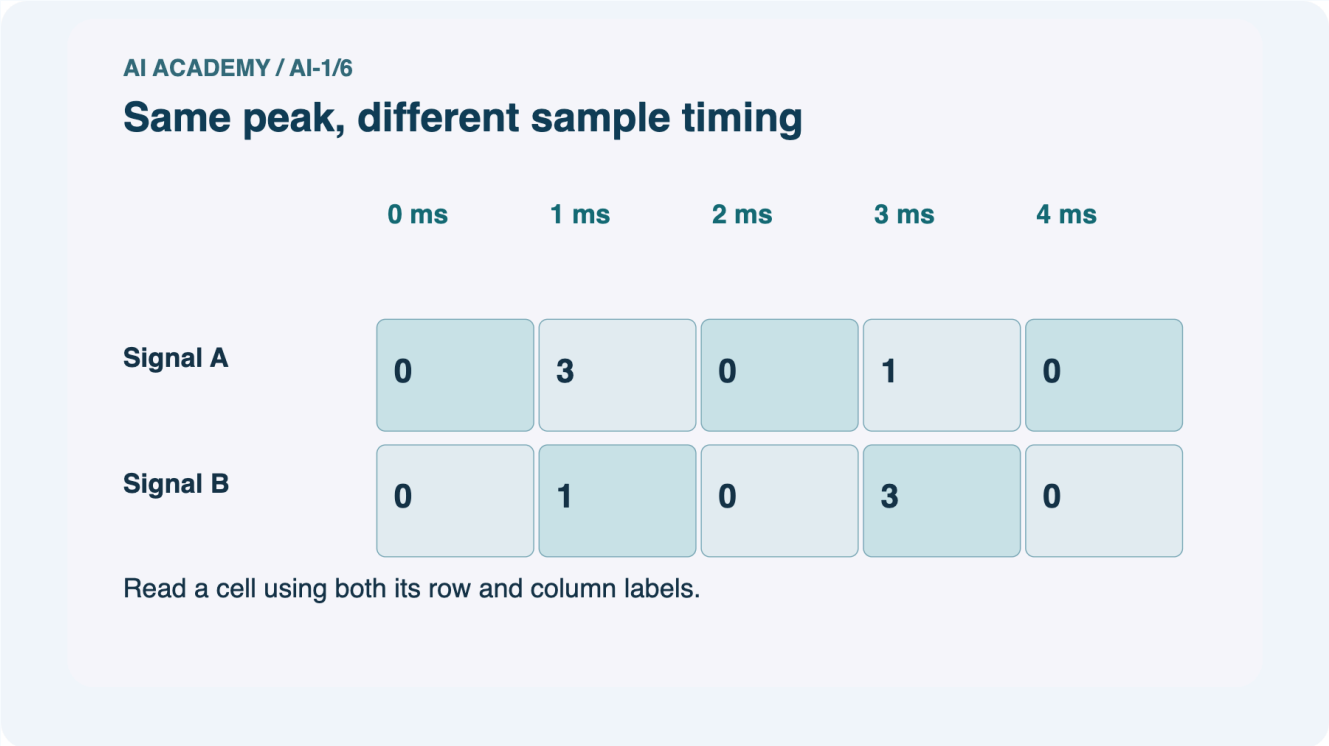


Follow the labelled picture with the matching explanation. Open the vector diagram

Timed sampling: Associate every measurement with its sampling time. → **Signal** A: Its largest **sample** 3 occurs at 1 ms. → **Signal** B: Its largest **sample** 3 occurs at 3 ms. → Maximum summary: Both maxima are 3, but the strongest recorded tap arrives at different times.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Same peak, different sample timing

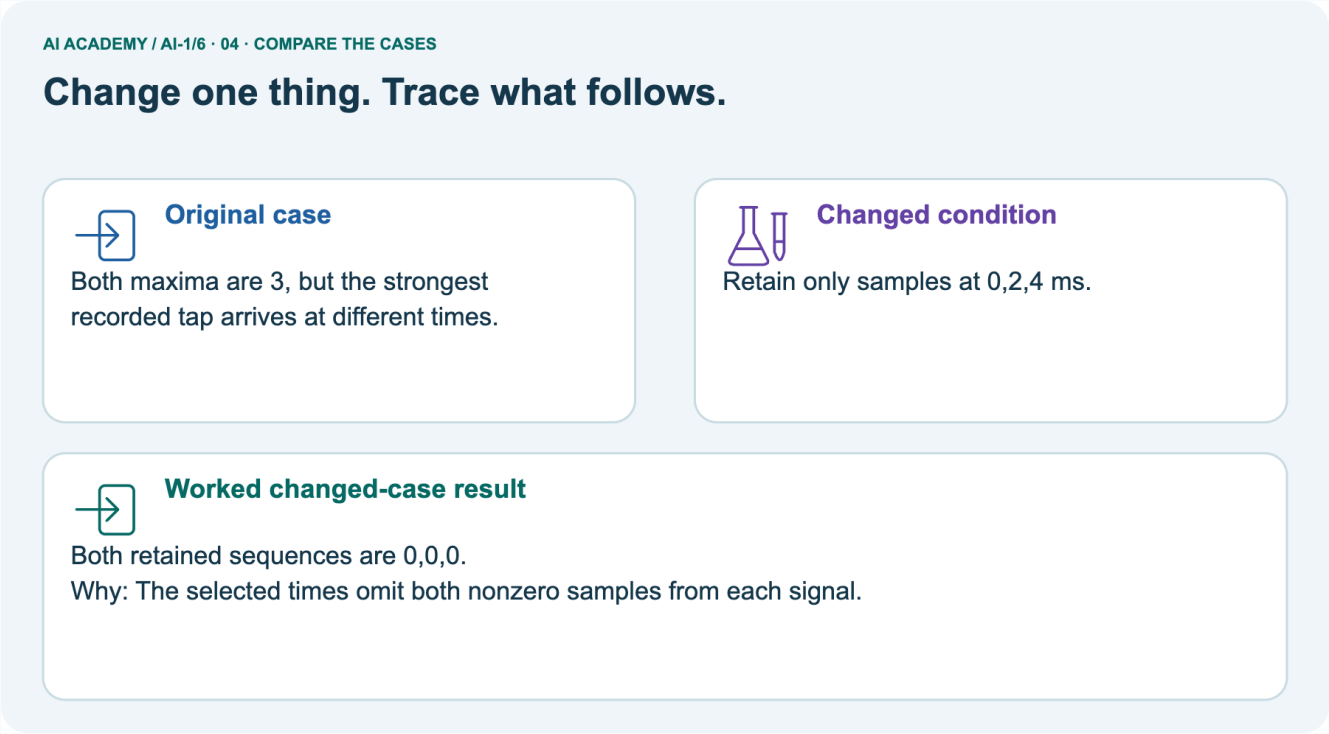


Follow the labelled picture with the matching explanation. Open the vector diagram

Same peak, different **sample** timing. **Signal** A: 0 ms = 0, 1 ms = 3, 2 ms = 0, 3 ms = 1, 4 ms = 0. **Signal** B: 0 ms = 0, 1 ms = 1, 2 ms = 0, 3 ms = 3, 4 ms = 0.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



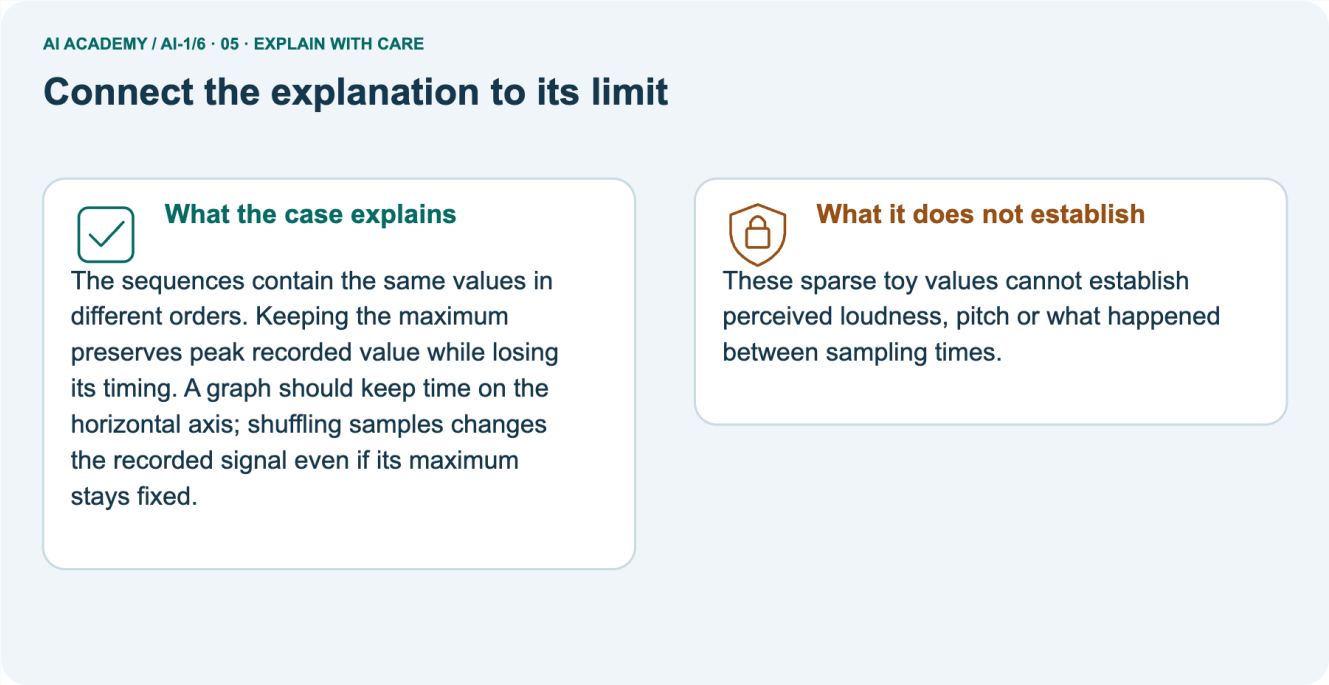
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both maxima are 3, but the strongest recorded tap arrives at different times. Changed condition: Retain only **sample**s at 0,2,4 ms. Expected result: Both retained sequences are 0,0,0.

Reason: The selected times omit both nonzero **sample**s from each **signal**.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

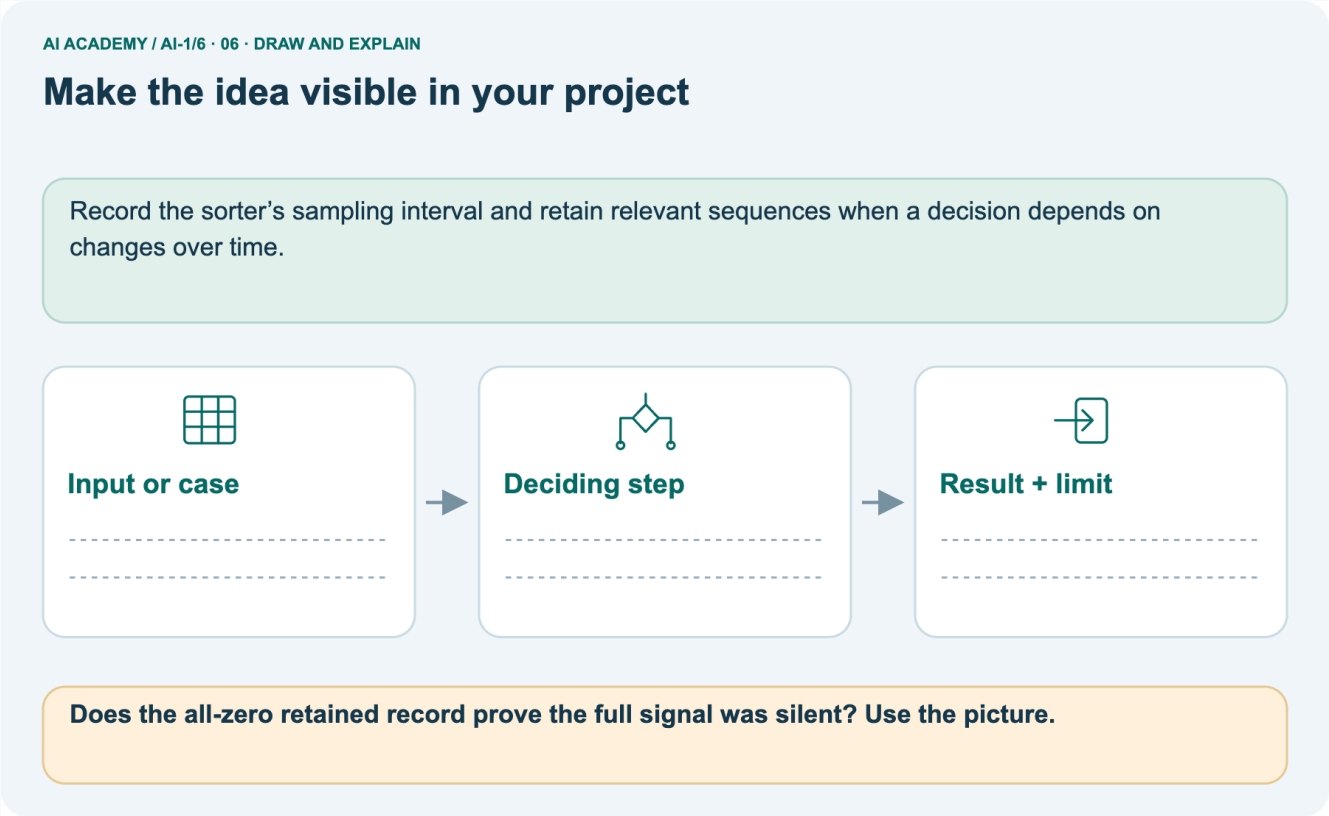
The sequences contain the same values in different orders. Keeping the maximum preserves peak recorded value while losing its timing.

A graph should keep time on the horizontal axis; shuffling **sample**s changes the recorded **signal** even if its maximum stays fixed.

Limit: These sparse toy values cannot establish perceived loudness, pitch or what happened between sampling times.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Record the sorter's **sampling interval** and retain relevant sequences when a decision depends on changes over time. Draw your **input**, deciding step, result and limit.

Supported question: Does the all-zero retained record prove the full **signal** was silent? Use the picture.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.